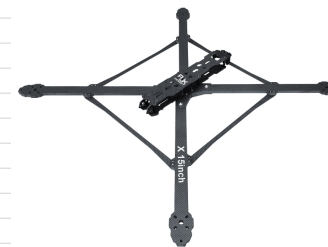
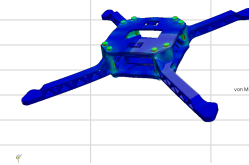
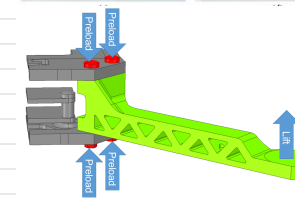
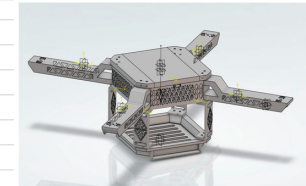
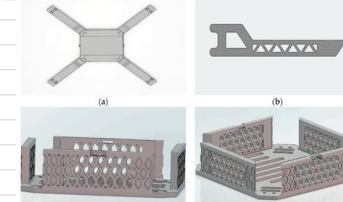
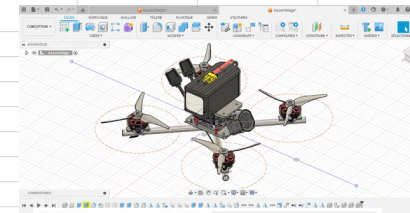


	Drone Arm Materials						
	Name	Density	Young Modulus	Poisson's Ratio	Yield Strength	Avg Cost	notes:
	Material	rho_kg_m3	E_Pa	nu	Pa	Cost_USD_per_m	
	Carbon Fiber Composite (CFRP)	1,600.00	7.00E+10	0.3	600,000,000.00	\$30.00	very light, strong, high stiffness
	Aluminum Alloy	2,700.00	6.90E+10	0.33	275,000,000.00	\$7.50	strong, durable, moderate weight
	Fiberglass Composite (GFRP)	1,900.00	2.50E+10	0.28	300,000,000.00	\$15.00	medium weight and stiffness
	PLA Plastic	1,250.00	3.50E+09	0.36	60,000,000.00	\$3.50	easy to 3d print but low structural strength
	ABS Plastic	1,050.00	2.00E+09	0.35	40,000,000.00	\$3.50	more impact resistance then pla
	Wood (Birch)	600	1.00E+10	0.35	80,000,000.00	\$2.00	low cost, heavier, not consistent
	Weight of Each Drone Component						
	Name	Individual Weight	Total Weight				
	Battery	0.3kg	0.3kg				
	Each Motor, 4 Motors	65g	0.26kg				
	Each Propeller, 4 Propellers	10g	0.04kg				
	Each Electronic Speed Controller,	30g	0.12kg				
	Flight Controller and Electronics	0.05kg	0.05kg				
	Receiver/GPA	0.03kg	0.03kg				
	Drone Frame (center body) exclud	0.2kg	0.2kg				
	Assumptions						
	Quadcopter drone with 4 motors and 4 propellers						
	Each motor can provide up to 1kg thrust						
	Accel. due to gravity: $g=9.81 \text{ m/s}^2$						
	Tot. weight of the drone components w/o the arms or the payload: 1kg						

Combination #	Design	Material
1	hollow circular tube	Carbon Fiber Composite (CFRP)
2	hollow circular tube	Aluminum Alloy
3	hollow circular tube	Fiberglass Composite (GFRP)
4	hollow circular tube	PLA Plastic
5	hollow circular tube	ABS Plastic
6	hollow circular tube	Wood (Birch)
7	Hollow Rectangular Beam	Carbon Fiber Composite (CFRP)
8	Hollow Rectangular Beam	Aluminum Alloy
9	Hollow Rectangular Beam	Fiberglass Composite (GFRP)
10	Hollow Rectangular Beam	PLA Plastic
11	Hollow Rectangular Beam	ABS Plastic
12	Hollow Rectangular Beam	Wood (Birch)
13	Triangle Truss Arm	Carbon Fiber Composite (CFRP)
14	Triangle Truss Arm	Aluminum Alloy
15	Triangle Truss Arm	Fiberglass Composite (GFRP)
16	Triangle Truss Arm	PLA Plastic
17	Triangle Truss Arm	ABS Plastic
18	Triangle Truss Arm	Wood (Birch)
19	I-Beam	Carbon Fiber Composite (CFRP)
20	I-Beam	Aluminum Alloy
21	I-Beam	Fiberglass Composite (GFRP)
22	I-Beam	PLA Plastic
23	I-Beam	ABS Plastic
24	I-Beam	Wood (Birch)
25	tapered hollow square tube	Carbon Fiber Composite (CFRP)
26	tapered hollow square tube	Aluminum Alloy
27	tapered hollow square tube	Fiberglass Composite (GFRP)
28	tapered hollow square tube	PLA Plastic
29	tapered hollow square tube	ABS Plastic
30	tapered hollow square tube	Wood (Birch)



RUX
HOBBY